

Fifth addendum to the 3.0.9/3.1.1 Release Notes

August 1st, 2025

Following a reorganization of the ITER SDCC cluster, it has become necessary to update the *master* 3.0.9 code version (and, as an accompaniment, update the 3.1.1 and 3.2.0 versions as well), to accommodate the new run environment. With this opportunity, a lot of other configuration updates for other HPC clusters in Member States' institutions, and various bug fixes as well as a few new features were also introduced. Please find below a brief description of these. For fuller details, refer to earlier User Forum debriefings and/or the code documentation and repository logs. I have also retroactively tagged the other *master* version updates to find the particular commits at which they happened more easily. The new tags have a lowercase letter added to the version number to indicate which update they are. Thus, the update discussed here will be tagged as 3.0.9e and 3.1.1e, as they are the fifth ones since the release of those code versions. The **Eirene** MsV version it links to carries the number 1.2.0, now covered by a CC_BY-NC-ND licence.

Among the changes imported with these updates, there are three that will affect the default behaviour of most standard SOLPS-ITER runs and required rebuilds of some of the CI test cases, although the overall impact on the solution was always small:

1. The handling of the time census stratum upon a run (re)start has been made simpler, by which the code will automatically fill in a neutral background if needed. This is done by a reassignment of the default value of the NTMSTP variable from block 13 of the **Eirene** input file depending on the other time census stratum settings. The time census weights are also now properly rescaled if the run restart involves a change in timestep. It is also now properly broadcast for cases using MPI parallelization. Thanks to [Reiter Detlev EXT](#) for his help in solving this matter.
2. A bug was discovered by [Van Uytven Wim EXT](#) in the treatment of some heavy-heavy collisions in **Eirene** which affected the calculation of the associated momentum sources. [Reiter Detlev EXT](#) promptly provided a fix, which makes the solution for coupled runs with molecule collisions evolve somewhat.
3. [Van Uytven Wim EXT](#) discovered a different bug in the calculation of the flux limit for the electron velocity associated with the thermo-electric current, which affects runs where *calf* is non-zero and the potential equation is being solved.

Most of the changes are related to environment updates and configuration file fixes, such as:

- Modified ITER configuration files, now using IMAS modules with DD-4.0.0, AL-5.4.x, IDStools/2.2.0, GGD/1.13.0, and AMNS/1.6.0, and job submission scripts with the new partition names.
- A streamlining of the job submission template files, so that separate files for coupled runs and/or compressed output are no longer necessary, proposed by [Zito Antonello EXT](#).
- [Coster David EXT](#) has updated the information for usage of the *solps-mdsplus.ipp.mpg.de* MDSplus SOLPS database server maintained at IPP, which is now back in operation.
- A first set of configuration files for the PITAGORA machine, still missing submission scripts until it goes in full operation, provided by [Coster David EXT](#), as well as clean-up of the ITM Gateway configuration files.
- Updated configuration files for the IPP cluster, as well as the new MPCDF one, both in Garching, provided by [Coster David EXT](#) and [Zito Antonello EXT](#).
- Updated configuration files for the Ljubljana cluster (UL), provided by [Kos Leon EXT](#).
- A *darwinsubmit* script for MacOS systems, provided by [Zito Antonello EXT](#).
- Adaptations to IMAS environment updates, including version 4.0.0 of the Data Dictionary, Access Layer 5, and compatibility with the **IMAS-Paraview** visualization tool.
- Safeties against the use of the *ifx* compiler, which was shown to be unable to run OpenMP parallelized **B2.5** cases or the **Uinp** case build-up program. It is still hoped that these compiler bugs will have been fixed in the 2025 API version, but that remains to be tested.

Then, there are some additional features that have been brought in from the *develop* branch after it was deemed they were mature enough:

- The new IMPROVED MPI parallelization strategy for **Eirene**, developed by [Kawamura Gakushi EXT](#).
- Support for use of the OPEN_MPI library when parallelizing **Eirene**.
- Support for hybrid OpenMP+MPI parallelization of **Eirene**, thanks to bug fixes from [Saxena Gaurav EXT](#).
- Support for both fixed- and JSON-formatted **Eirene** input files in the various job management scripts, by relying on the symbolic link *fort.1* instead of *input.dat*.
- Two new feedback schemes, NA_FEEDBACK_CHOICE = 17 and 18, for the total radiation from a given isonuclear sequence in a rectangle of cells or the zone outside the separatrix, respectively, as well as two new rescaling modes, NA_FEEDBACK_OPTION = 8 or 9, to recover some of the possibilities present in SOLPS4 (provided by [Pshenov Andrei](#)).
- The new formulation of the velocity-dependent stress-viscosity tensor in the Zhdanov closure, provided by [Makarov Sergei EXT](#), accessible with the switch setting *b2tral_zh_vis_type.eq.2*, as well as the associated alternative formulation of the Luciani ion heat flux limit, now valid for plasmas beyond hydrogen-majority ones (governed by the *b2trcl_luc_et_form* switch), and inclusion of the heat flux due to the velocity-dependence of the stress viscosity tensor in the *Balance* Matlab scripts (following a report from [Zito Antonello EXT](#)).
- An optional stabilization term for the BCCON=14 drift-compatible sheath density boundary condition, governed by the *b2stbc_stab_coeff_sheath_ni* switch (provided by the St. Petersburg group).
- Additional tracing output quantities and corrections in *user.trc* related to ballooning and use of a target shaping factor when computing heat loads, and total particles contents in *intshrt.trc* (provided by [Pshenov Andrei](#)).
- Revival of the *b2movies.nc* output option (suggested by [Zito Antonello EXT](#)).
- Extension of the NFLAGS switch capability in **Eirene** to obtain connectivity information tables for triangular grids (provided by the FZJ group).
- A sanity check in the *carre* script to verify that the *carre.dat* file, if produced, does contain grid data before proceeding.
- New capabilities in the post-processing analysis scripts, such as *last_2d* and *plot_trc*, provided by [Pshenov Andrei](#).
- Flushing of **B2.5** output files immediately after they are written to get all possible output in case of a crash.
- Reduced number of "duplicate element" warning messages from MA28 (requested by [Makarov Sergei EXT](#)).
- Possibility to request an exclusive node allocation when submitting a SLURM job, using a *-x* option.
- An updated *easybuild-local.sh* script for code installation, provided by [Kos Leon EXT](#).
- An automated way to install several of the dependency libraries (MSCL, GR, and GKS), provided by [Zito Antonello EXT](#).
- Some output corrections and prettifying.
- Lots of updates and clarifications to the code comments and documentation, following suggestions from users and reflecting the changes described here.

Some other notable miscellaneous bug fixes:

- Bug fixes for restarting and continuing a case with a *fort.44* file written with a different format and/or after the time census stratum has been added or removed.
- Bug fix to avoid a restart effect with some feedback schemes (provided by [Pshenov Andrei](#)).
- Bug fixes from [Horsten Niels EXT](#) for the accounting of the radiation from heavy-heavy collisions in **Eirene**.
- Bug fix proposed by [Wu Haosheng EXT](#) for the computation of the electron convective heat flux tally (qconvexreg) to properly account for the parallel current.
- Bug fixes to particle balance calculation in *blnc.trc* tracing file (provided by [Pshenov Andrei](#) and [Zito Antonello EXT](#)).
- Bug fix from [Wu Haosheng EXT](#) for the repartition of radial surfaces in **Carre** when adapting the grid to a divertor target contour.
- Bug fix from [Wu Haosheng EXT](#) to correct the back-mapping of **Eirene** data onto the **B2.5** grid for snowflake cases.
- Bug fix for the dimension of external source arrays (reported by [Jung Dami EXT](#)).
- Modified sanity check for boundary condition cells in 1-D cases (requested by [Coster David EXT](#)).
- Correcting **Uinp** to provide a working **Eirene** input file in the case where no TRIM A_on_B files are listed.
- Bug fix to **b2plot** user option to obtain the CII radiation.
- Bug fix to the **dg2ef** converter program (from a **DivGeo** to an EQDSK equilibrium format) (reported by [Korving Sven](#)).
- Bug fixes for the total ion density in the *read_b2fplasmf* Matlab script and for limiter cases in *particle_balance.py* (reported by [Tonello Elena EXT](#)).

To obtain the new 3.0.9 *master* version, use the *solps-iter_update* script. Be mindful that, because many of the setup and configuration files change with this update, the code will most likely fail to compile the first time around. In that case, close your session, restart it to load the updated setup and configuration files, and attempt the compilation again. The 3.0.9 *develop* branch has been rebased above the updated *master*, so if you need to update that one, use the *solps-iter_update_develop* script instead. To check if the update was successful, you can use the *print_versions* script, and all five code components should indicate a GIT hash starting with "3.0.9e".